Block 5 – SDN & Cloud-Native Networking

SDN, NFV, Overlays & Network Automation

Arquitectura de Xarxes

Universitat Pompeu Fabra

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Outline

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What If You Could Program Your Entire Network?

How are modern networks built and automated under the hood?

Traditional networks are configured device by device.
What if the network was as programmable as software?

Learning objectives:

1. Explain SDN architecture and why centralized control matters
2. Understand NFV and how it replaces hardware appliances
3. Describe overlay networks (VXLAN) and why they scale beyond VLANs
4. Define cloud-native design and explain how containers, microservices, and IaC work together

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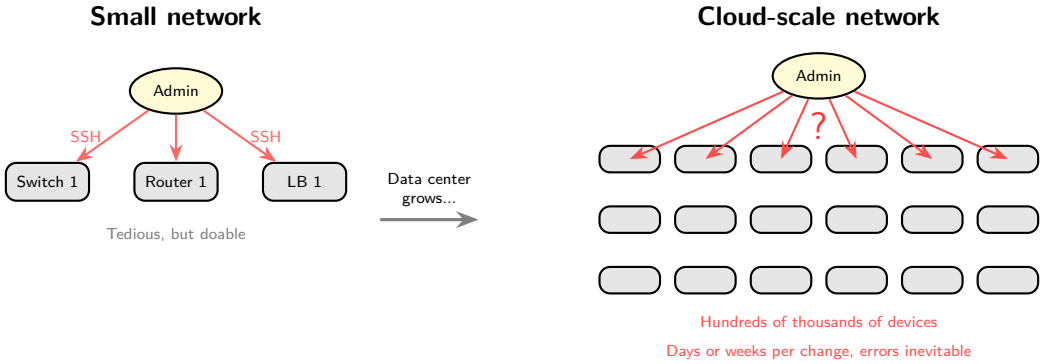
The Problem with Traditional Networks



At cloud scale, manual configuration breaks completely. There has to be a better way.

Discussion: The Cost of Manual Networks

Why have traditional networks survived so long despite their scalability problems?

- Hint: think about reliability, vendor relationships, and risk aversion
- What would it take for an organization to change?

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Software-Defined Networking (SDN)

Software-Defined Networking is an approach to network management that makes the network **programmable** by separating the logic that controls traffic from the hardware that forwards it.

- **Separate the control plane from the data plane**: the control plane decides where traffic goes; the data plane just forwards it. SDN puts them in separate layers
- **Centralize control**: one (or a cluster of) software controller(s) has a global view of the network
- **Use software to configure devices**: no manual CLI per device; push rules programmatically
- **Open interfaces**: any application can interact with the network via standard APIs

The network becomes as flexible as software: you can change how traffic flows without touching hardware.

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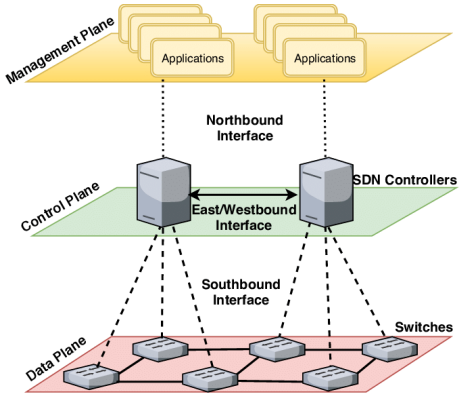
SDN: The Centralized Brain

- The controller is the **brain** of the network: global view, all forwarding decisions
- Maintains a real-time **topology database**
- **Programmable via API**: scripts and apps push rules without touching the CLI
- Multiple controllers sync for redundancy [2]

Open-source: OpenDaylight, ONOS
Proprietary: Cisco ACI, VMware NSX



[2] Latif et al., "A Comprehensive Survey of Interface Protocols for SDN," arXiv:1902.07913, 2019.

How Does the Controller Talk to Switches?

The controller needs a **common language** to program any switch, regardless of vendor. That language is called the **southbound protocol**.

- **OpenFlow** (Stanford, 2008): the first standard protocol [1]
  - Controller sends **flow rules**: "if packet matches X, do Y"
  - Proved the concept works on real hardware
  - Rarely used in production today
- **Modern alternatives**: gNMI, NETCONF/YANG, vendor APIs (Cisco ACI, VMware NSX)

Think of it as a universal remote control: one controller, any switch brand.

[1] ONF, "Software-Defined Networking: The New Norm for Networks," ONF White Paper, 2012.

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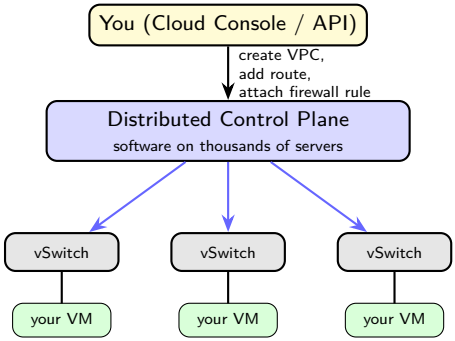
SDN Today: It Is Everywhere

Every time you create a network in AWS, Azure, or GCP, **SDN is doing the work**. No engineer is SSHing into a switch.

- **Cloud providers** (AWS, Azure, GCP): SDN logic distributed across every server in the data center
- **Telcos**: network functions running as **containers**, controlled via software APIs

The SDN **principles** are identical to what we studied:

- Separate decisions from forwarding
- Control via software and APIs



Discussion: SDN Trade-offs

If the SDN controller fails,
what happens to the network?

- Hint: single point of failure vs distributed control
- How do real deployments address this? (clustering, failover)

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Network Functions Virtualization (NFV)

A **network function (NF)** is any processing task a device performs on traffic: routing, filtering, load balancing, intrusion detection. Traditionally, each NF ran on a dedicated hardware appliance. NFV moves them to software on commodity servers.

- Traditional appliances: expensive (€10K–€100K+), slow to procure, hard to scale
- **VNF** (Virtual Network Function): an NF running on top of a VM [1]
- **CNF** (Cloud-Native Network Function): an NF running on top of a container
- Can be instantiated, scaled, or deleted in minutes via software

SDN decides where traffic goes. NFV decides what happens to it.

[1] ETSI GS NFV 002, "NFV Architectural Framework," v1.2.1, 2014.

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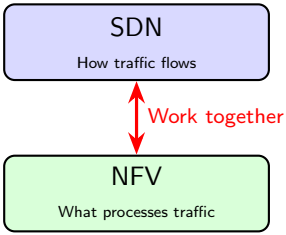
NFV + Cloud Integration

- Deploy VNFs on cloud infrastructure → **minutes** instead of months
- **Scale horizontally**: add more instances under load
- **Update easily**: software upgrades, no truck rolls
- **Cost reduction**: commodity servers vs specialized hardware

No hardware procurement, no shipping, no rack-and-stack.

SDN + NFV: Complementary Technologies



Example: HTTP request entering a data center

1. SDN controller sees incoming traffic
2. Forwards it to a **VNF firewall** (NFV) for inspection
3. Firewall approves; SDN routes it to the **VNF load balancer**
4. Load balancer distributes to backend servers

- SDN **steers** traffic (forwarding rules)
- NFV **processes** traffic (filter, balance, inspect)

Discussion: When Hardware Still Wins

Can you think of a scenario where a physical appliance is better than a VNF?

- Hint: think about latency, throughput, and specialized workloads
- What about hardware accelerators (FPGAs, SmartNICs)?

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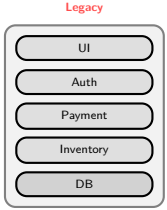
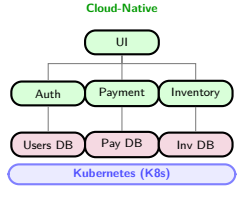
What Does Cloud-Native Mean?

- SDN and NFV give us programmable, virtualized infrastructure
- But applications can still be designed the **old way**: one big monolith, manually deployed, that happens to run in the cloud
- **Cloud-native** is the design philosophy that fully exploits what SDN, NFV, and containers make possible
- Key question: are you just *lifting and shifting* an old app, or designing for the cloud from the start?

	Legacy	**Cloud-Native**
Design	Monolith in a VM	Microservices
Scaling	Manual / vertical	Auto / horizontal
Failure	Restart whole app	Isolate service
Deploy	Manual steps	CI/CD pipeline





Cloud-Native Design Principles

Cloud-native applications are built around a small set of principles: [1]

- **Microservices**: one responsibility per service; independent deploy and scale
- **Containers**: standard, portable packaging; same image from laptop to production
- **Declarative configuration**: describe *what* you want, not *how* to achieve it
- **Immutable infrastructure**: never patch a running instance; replace it with a new image
- **Automated delivery**: CI/CD pipelines build, test, and deploy without manual steps

[1] Burns et al., *Kubernetes: Up and Running*, 3rd ed., O'Reilly, 2022.

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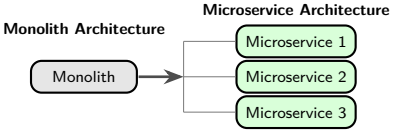
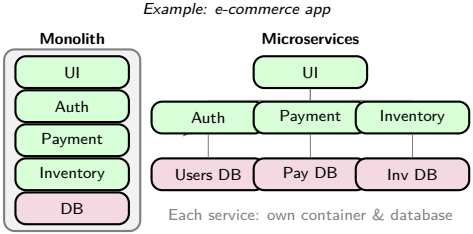
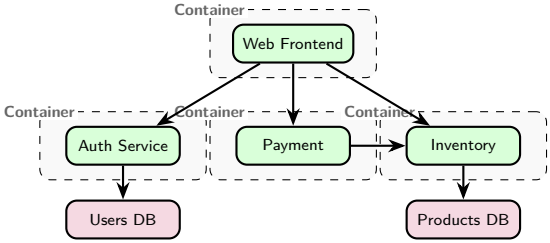
Microservices: From Monolith to Distributed

- **Monolithic** app: one big process, one deployment, one codebase
- **Microservices**: app split into small, independent services, each in its own container
- Rule of thumb: **1 microservice = 1 container**





Microservices: Network Challenges



- Services communicate over the **network** (HTTP/REST, gRPC)
- Network implications for hundreds of containers:
  - **Service discovery**: "where is the payment service?"
  - **Load balancing**: distribute requests across replicas
  - **Observability**: trace requests across services
- Every arrow = **network call** → latency and failure risk
- If Inventory is down → Payment and Web are affected (**cascading failure**)
- Solutions: retries, timeouts, circuit breakers, service meshes

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Kubernetes: The Cloud-Native Platform

These principles require a platform that handles scheduling, networking, scaling, and self-healing automatically. That platform is **Kubernetes (K8s)**, introduced in Block 4 Session 2.

From a networking perspective, K8s provides:

- Each **Pod** gets its own IP address (dynamic, managed by the platform)
- **Services** provide stable DNS names regardless of which host Pods run on
- **Ingress** controllers expose services externally with L7 routing
- **Network Policies** define which Pods can talk to which (micro-segmentation)
- **Container Network Interface (CNI)** plugins handle the overlay wiring (typically VXLAN-based)

Cloud-native networking requirements (dynamic IPs, cross-host comms, load balancing, policies) are exactly what K8s was built to solve.

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Discussion: Is Cloud-Native Always Better?

A bank has a 15-year-old monolithic payment application running on two dedicated servers.
Should they rewrite it as cloud-native microservices?

- Hint: think about cost, risk, team skills, and whether the current system actually has a scaling problem
- Is "cloud-native" a tool for specific problems, or always the right answer?

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Why Overlay Networks?

- SDN and NFV give us programmable, virtualized infrastructure
- Multiple tenants share the same physical network: their traffic must be **isolated**
- We know VLANs (Block 3), but VLANs have hard limits:
  - **12-bit ID**: maximum 4,096 networks
  - **Layer 2 only**: cannot span Layer 3 boundaries
  - Moving a VM to another host may require VLAN reconfiguration
- A cloud provider hosting millions of tenants cannot rely on VLANs
- **Overlay networks** solve this: scalable isolation that works across any physical topology

We need a technology that scales beyond 4K networks and works across L3 boundaries.

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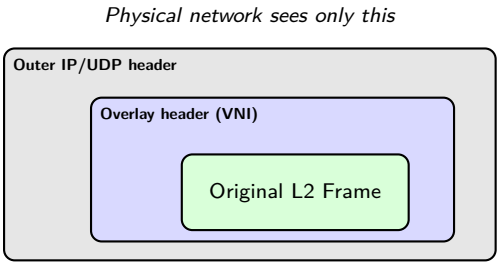
Overlay Networks: Concept

- An **overlay network** is a virtual network built **on top of** the physical one
- Uses **encapsulation**: the original frame is wrapped inside a physical packet
- The physical network only sees the **outer headers**: tenant traffic is invisible
- Each tenant gets **full isolation** without the physical network knowing about them



VXLAN: Overview

- **VXLAN** (Virtual eXtensible LAN): most widely used overlay protocol
- Encapsulates L2 frames in **UDP packets** over the physical network
- Uses a **24-bit VNI** (VXLAN Network Identifier)
  - $2^{24} =$ **16 million** virtual networks (vs 4,096 VLANs)
- Endpoints: **VTEPs** (VXLAN Tunnel Endpoints) [1]
  - Encapsulate at source, decapsulate at destination

[1] IETF RFC 7348, "Virtual eXtensible Local Area Network (VXLAN)," 2014.

VXLAN: How It Works

- VM A sends a frame → VTEP 1 wraps it in UDP with a VNI
- Travels over the physical network as a regular UDP packet
- VTEP 2 strips outer headers, delivers original frame to VM B
- VMs on the **same VNI** form an isolated L2 domain

Overlay Benefits for Scalability

- **16 million virtual networks** (vs 4,096 VLANs)
- Physical network does not need to know about tenants
- VMs can **migrate** across hosts without reconfiguring the underlay
- Decouples **virtual topology** from physical topology [1]

[1] IETF RFC 7348, "Virtual eXtensible Local Area Network (VXLAN)," 2014.

Discussion: Overlay Networks

Why can the physical network remain simple
if we use overlay networks?
What are the trade-offs of encapsulation?

Hints: think about overhead (extra headers), Maximum Transmission Unit (MTU) implications, and troubleshooting complexity.

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Why Container Networking?

- Overlay networks give us scalable, isolated virtual networks across any physical infrastructure
- Modern applications are split into **dozens of independent services**, each deployed as a container
- Each container needs its own **network identity** (IP address, hostname)
- Containers must reach each other, possibly on **different physical hosts**
- The overlay networks we just studied are exactly what makes cross-host container communication possible at scale

How does networking work inside and between container hosts?

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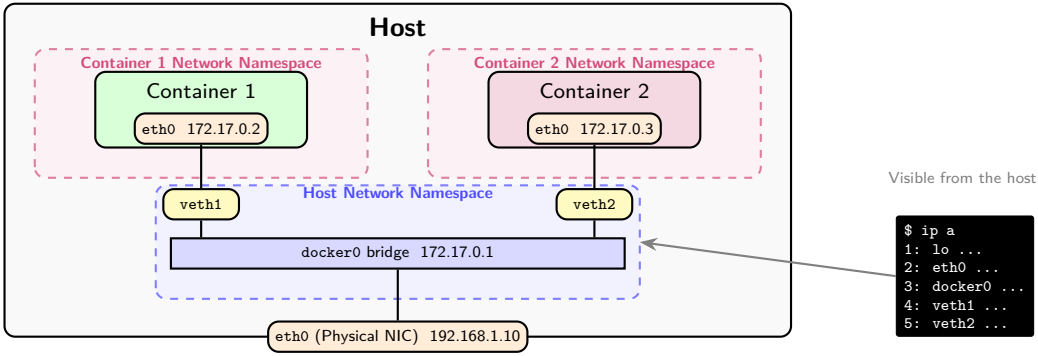
Container Networking: Starting Point

- Every container gets its own **network namespace**: isolated IP stack, interfaces, routing table
  - Network namespace: private isolated environment where the container sees only its own interfaces and routes
- Docker wires containers via **virtual bridges** and **veth pairs** (virtual cable)
- Containers reach each other, the host, and the internet depending on the **network model**

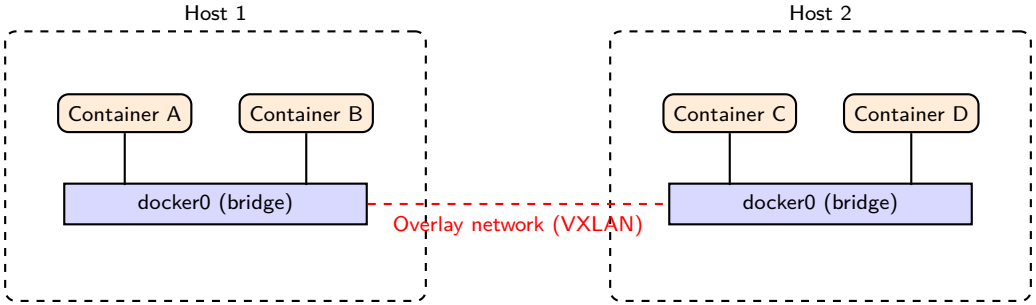


Container Network Models

- **Bridge network (default):**
  - Containers on the same host connected via a **virtual bridge** (`docker0`) [1]
  - Each container gets a **veth pair** (virtual Ethernet interface)
  - **NAT** for external traffic; containers share host's IP
- **Host network:**
  - Container uses the **host's network stack** directly
  - No network isolation, but **no NAT overhead**
  - Use for performance-critical applications
- **Overlay network:**
  - Containers on **different hosts** connected via VXLAN overlay
  - Same overlay concept we just covered, now applied to containers
  - Enables **multi-host** container deployments

[1] Docker, Inc., "Docker Networking Documentation," docs.docker.com/network.

Container Networking: Visual



- Within a host: containers use a **bridge** network
- Across hosts: an **overlay** connects the bridges (VXLAN from the previous section)

Discussion: Containers vs VMs

If containers are faster and lighter than VMs,
why do companies still use VMs?

- Hint: security isolation, legacy applications, compliance requirements
- In practice, containers often run inside VMs

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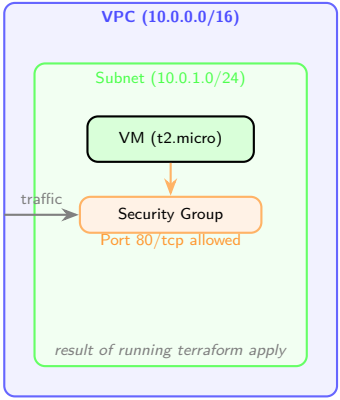
Infrastructure as Code (IaC)

- SDN, NFV, overlays, and cloud-native containers are all programmable: if still configured configured **by hand**, we are back to slow, error-prone, and unreproducible
- **IaC**: manage infrastructure through **code files**, not manual clicks or dashboards
- Describe the desired state in a file $\rightarrow$ tools apply it automatically
- **Reproducibility**: same config $\rightarrow$ same infrastructure, every time
- **Version control**: track changes in Git, review in pull requests, rollback instantly
- **Speed**: deploy entire environments in minutes [1]

[1] Morris, *Infrastructure as Code*, 2nd ed., O'Reilly, 2021.

IaC Example: Terraform (Conceptual)

```
# Define a VPC [1]
resource "aws_vpc" "main" {
  cidr_block = "10.0.0.0/16"
}
# Define a subnet
resource "aws_subnet" "public" {
  vpc_id   = aws_vpc.main.id
  cidr_block = "10.0.1.0/24"
}
# Define a VM (EC2 instance)
resource "aws_instance" "web" {
  ami     = "ami-0abcdef1234567890"
  instance_type = "t2.micro"
  subnet_id = aws_subnet.public.id
}
# Define a security group
resource "aws_security_group" "web" {
  vpc_id = aws_vpc.main.id
  ingress {
    from_port = 80
    to_port = 80
    protocol = "tcp"
  }
}
```



[1] HashiCorp, "Terraform Documentation," terraform.io/docs.

Discussion: IaC Risks

If infrastructure is defined as code,
what happens when someone pushes a bug?

- Hint: code review, staging environments, automated testing
- How is this similar to software development best practices?

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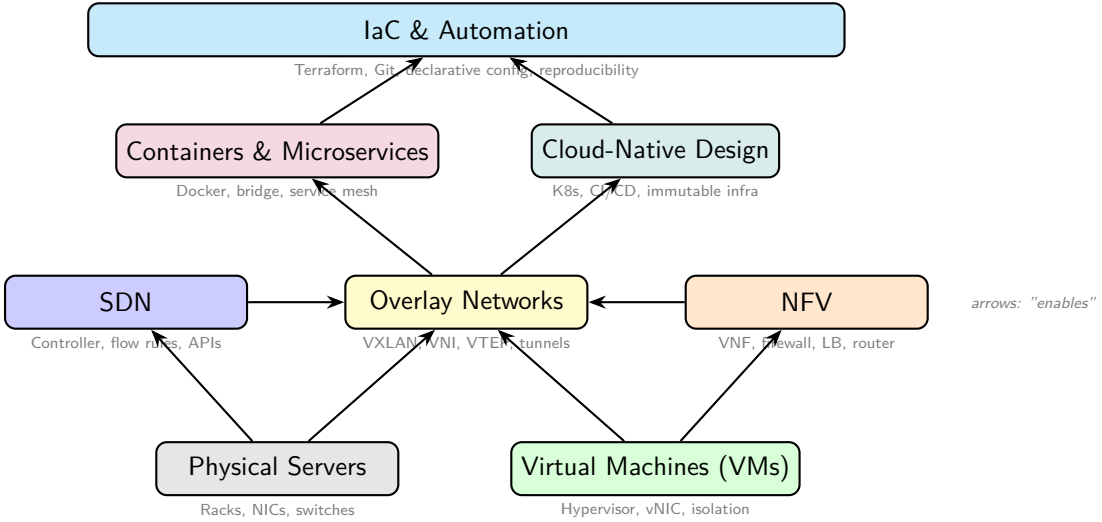
The Full Picture: Where Everything Fits



Each layer builds on the previous one: from bare metal to fully automated cloud-native infrastructure.

Key Takeaways

1. Traditional networks are **manual, rigid, and do not scale**
2. **SDN** separates control from data plane: centralized, programmable, API-driven
3. **NFV** replaces hardware appliances with software: flexible, fast to deploy, cheap
4. SDN and NFV are **complementary**: SDN steers traffic, NFV processes it
5. **VXLAN** overlays scale isolation to 16 million networks (vs 4,096 VLANs)
6. Containers use **bridge networks** on a single host and **overlay networks** across hosts
7. **Cloud-native** means designed for the cloud: microservices, containers, declarative config, immutable infra, automated delivery
8. **Microservices** require service discovery, load balancing, and resilience mechanisms (circuit breakers)
9. **IaC** automates infrastructure: reproducible, version-controlled, auditable

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Discussion

A company runs 200 microservices in containers.
One Friday at 5 PM, a manual network change
breaks communication for 50 of them.
How could SDN and IaC have prevented this?

Hints: centralized control, automated testing, instant rollback, reproducibility.

References

1. McKeown et al., "OpenFlow: Enabling Innovation in Campus Networks," *ACM SIGCOMM CCR*, vol. 38, no. 2, 2008.
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